FoodFlow Analytics

Excel Phase — Data Preparation & Validation Report

1. Project Overview

Project: Food Delivery Operations & Revenue Intelligence
Current phase: Excel data preparation, validation and feature engineering.
Dataset: 60,000 synthetic food-delivery orders modeled on a Zomato/Swiggy-style business scenario. It is not proprietary platform data.

2. Excel Phase Objective

The Excel phase converts the raw order transactions into an analysis-ready table. We preserved the raw data, profiled the dataset, checked data quality and financial consistency, and created analytical columns that will later support SQL, Python and Power BI.

3. Workbook Structure

Worksheet	Purpose
Orders_Raw	Original 60,000-order dataset; preserved without modification.
Orders_Cle	Working dataset with validation and analytical columns.
Data_Dictionary	Field definitions and business meanings.
Validation	Dataset-level quality checks.
Financial_Validation	Order-total calculation validation.

4. Dataset Profile

Metric	Result
Orders	60,000
Original columns	17
Date range	01-Jan-2025 to 31-Dec-2025
Missing values	0 in the inspected order fields
Duplicate order IDs	0
Platforms	Zomato and Swiggy

5. Data Validation Performed

• **Order ID uniqueness:** duplicate order IDs checked; result was 0.

• **Missing values:** checked the original order fields; result was 0.

• **Logical checks:** monetary fields were checked for negative values.

• **Delivery-time check:** promised delivery minutes were checked for invalid/non-positive values.

• **Financial consistency:** Calculated_Total was created from subtotal – discount + delivery fee + packaging fee + tax.

• **Amount difference:** total_amount was compared with Calculated_Total; the visible validation showed 0 difference.

6. Financial Validation

Calculated_Total = subtotal – discount + delivery_fee + packaging_fee + tax

Amount_Difference = total_amount – Calculated_Total

This ensures the order-level financial calculation is internally consistent before further analysis.

7. Analytical Columns Created

Column	Purpose
Year	Year extracted from order_date.
Month	Numeric month for chronological analysis.
Month name	Readable month name.
Day name	Day-of-week analysis.
Hour	Hour extracted from order_time.
Meal Period	Breakfast, Lunch, Evening or Dinner.
Discount Percentage	Discount divided by subtotal.
Order Value Category	Low, Medium, High or Very High based on total_amount.
Delivery Slot	Morning, Lunch Peak, Afternoon, Evening Peak or Night.

8. Key Excel Logic

• Year: =YEAR(order_date)

• Month: =MONTH(order_date)

• Month Name: =TEXT(order_date,"mmmm")

• Day Name: =TEXT(order_date,"dddd")

• Hour: =HOUR(order_time)

• Meal Period: Hour < 12 = Breakfast; 12–15 = Lunch; 16–18 = Evening; 19+ = Dinner.

• Discount %: =IFERROR(discount/subtotal,0)

• Order Value Category: < ■250 Low; ■250–499.99 Medium; ■500–999.99 High; ■1000+ Very High.

9. Current Excel Deliverables

✓ Raw data preserved

✓ Data profiling/validation completed

✓ Financial consistency checked

✓ Logical checks completed

✓ Time-based analytical columns created

✓ Discount percentage created

✓ Order-value categories created

✓ Delivery-slot categories created

10. Next Phase — SQL

We will now move from Excel preparation to SQL. The next phase will create a relational database using the FoodFlow tables: orders, customers, restaurants, riders, food items, order items, delivery, cancellations, refunds, complaints, food waste and promotions. We will then perform business analysis using joins, aggregations, CASE, CTEs and window functions.

This report documents only the Excel phase; business conclusions and dashboard findings will be produced later from the SQL, Python and Power BI phases.